YOUCAM API SKIN AI HACKATHON · PROJECT DESCRIPTION

SkinWise AI

Your skin, understood. Then a smarter shelf.

An AI skincare companion that turns a real **YouCam AI skin analysis** into practical decisions: what your skin needs, what you already own, what is genuinely missing, and what it will cost at the lowest live price. Sometimes the best answer is: **you don't need to buy anything.**

YouCam Skin Analysis API v2 · AI routine · Shelf-first shopping · Live ₹ prices · Skin journey

THE PROBLEM

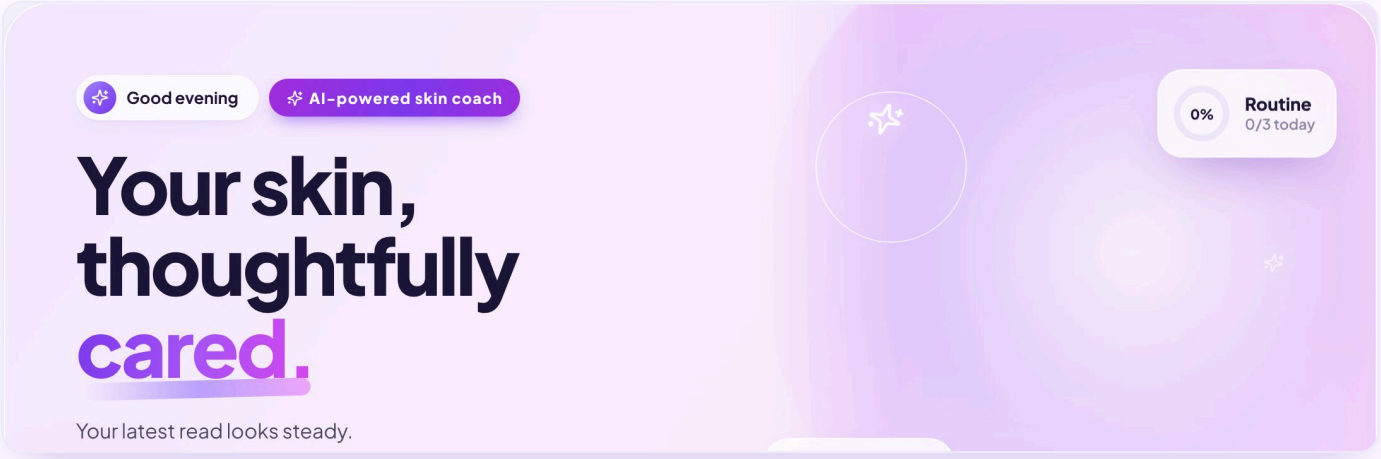
Skincare is guesswork and impulse buying

People choose products from ads and influencers, end up with duplicates and steps their skin does not need, and rarely know whether anything suits them. Skin-analysis apps stop at a score; shopping apps start with a sales pitch. Nothing connects the two honestly.

OUR SOLUTION

Analysis → routine → smarter shelf

SkinWise uses YouCam's real skin scores to build a personal AM/PM routine, checks the products the user already owns, and recommends only genuine gaps — real, in-stock products ranked for suitability and value within the user's budget, with direct links to the exact listing.



Overview — the latest YouCam read, today's routine and shelf coverage at a glance.

END-TO-END JOURNEY

01 · ANALYZE
AI skin scan
Selfie or photo → YouCam Skin Analysis API.

02 · UNDERSTAND
Skin profile
AI observations kept apart from user goals.

03 · ROUTINE
AI AM/PM ritual
Steps with reasons and priced product picks.

04 · SHOP SMARTER
Skin-to-Shelf
Owned first, gaps only, budget and lowest price.

05 · TRACK
Skin journey
Every scan saved and compared like-for-like.

For consumers

Understand skin in plain language, follow a simple routine, and spend less by using what they already own.

For retailers & brands

High-intent shoppers matched to products that fit a measured need — fewer regret purchases and returns.

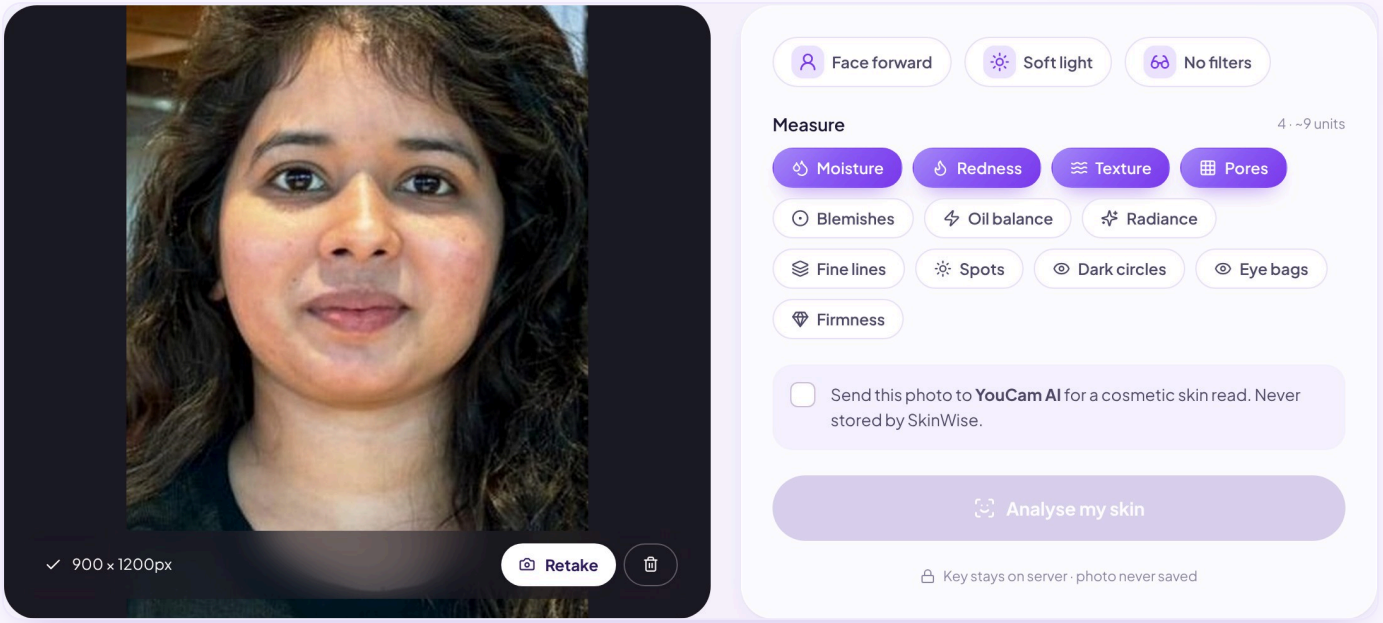
Built on trust

Consent per upload, selfies never stored, no invented scores, prices or products, no medical claims.

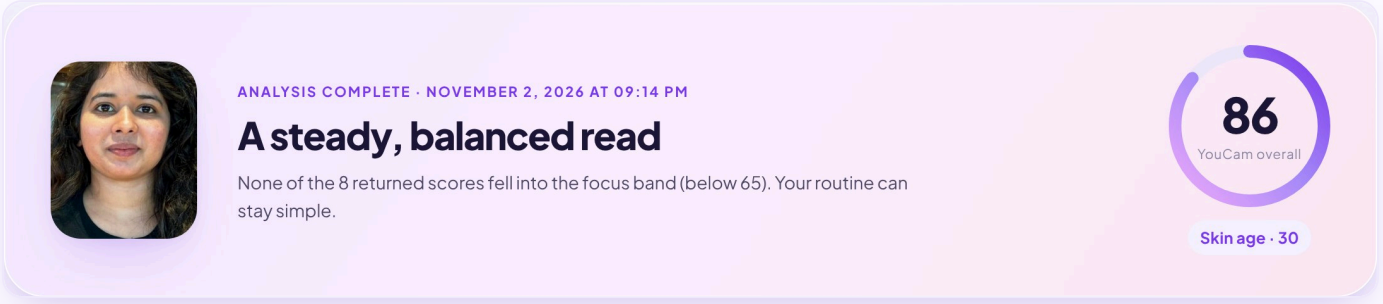
FEATURE 1 · AI SKIN ANALYSIS

A calm, private AI skin scan

Users take a live selfie (with a face-oval guide and automatic crop so the face fills the frame) or upload a photo, choose the concerns to measure, and give explicit consent. SkinWise validates the image before upload (JPG/PNG, ≤10 MB, shorter side ≥480 px), shows a live status while YouCam processes it, and resumes an in-progress scan after a page refresh.



Capture & consent — take a selfie or upload, pick concerns (each with an icon), consent, analyse.



Results — YouCam's overall score (86), skin age (30) and a plain-language summary. Real result from the analysed photo.

HOW SKINWISE USES THE YOUCAM API

1 Upload
`POST /s2s/v2.0/file`
Returns a signed URL; the selfie bytes are PUT from server memory — never written to disk.

2 Analyse
`POST /s2s/v2.0/task/skin-analysis`
SD concerns: moisture, redness, pores, texture, blemishes, oil balance, spots, dark circles…

3 Poll
`GET /task/skin-analysis/:task_id`
Status until success; results include ui_score, raw_score and mask URLs per concern.

4 Use
`overall · skin age · concerns`
Saved with a timestamp; powers profile, routine, Skin-to-Shelf and journey.

What the user sees

- **Overall score & skin age** returned by YouCam, plus a card for every concern with its 0–100 score, a plain-language band (Great / Fair / Focus), raw score and mask link.
- **Nothing estimated:** a requested metric that is not returned is labelled "Not returned".
- **Focus areas** (lowest scores) automatically shape the routine and shelf check.

Engineering & safety

- The browser talks only to our **Express backend**; the YouCam key stays server-side.
- Provider errors (face too small, too dark, no face, out of units) become **clear, actionable messages**.
- Each assessment stores internal ID, YouCam task ID, status, timestamps and normalized results.

Cosmetic insights only — SkinWise never diagnoses. Lighting, camera and angle can shift scores, and the app says so.

FEATURES 2–4 · OVERVIEW, SKIN PROFILE, AI ROUTINE

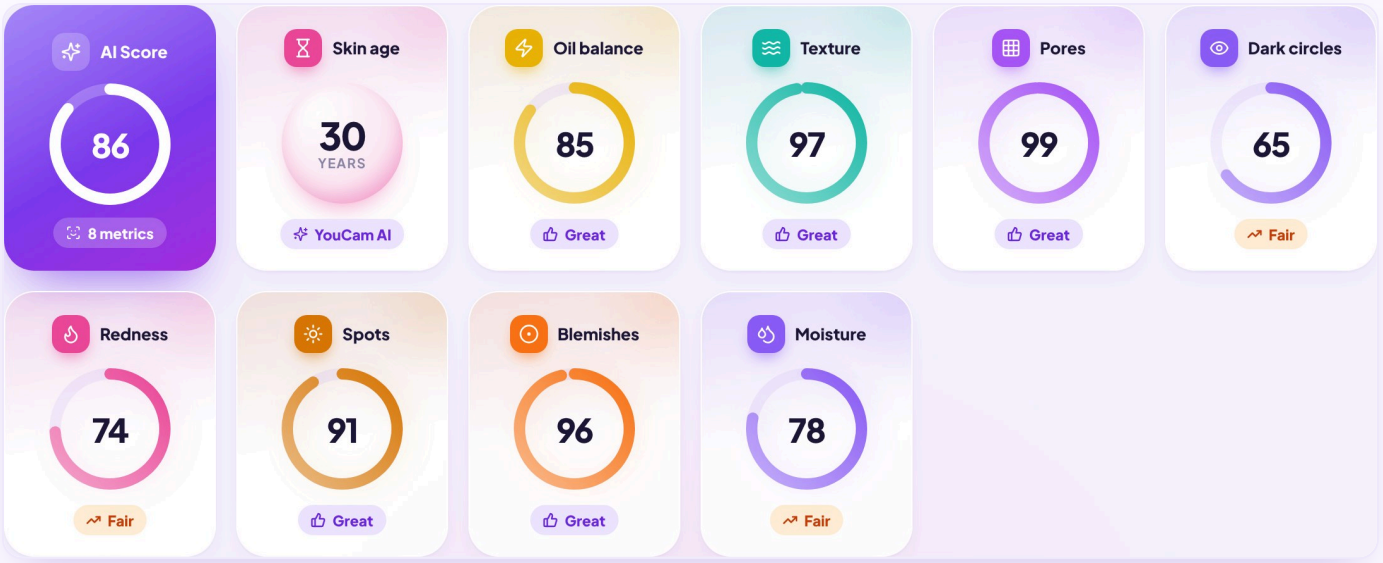
From scores to a daily ritual

Overview dashboard

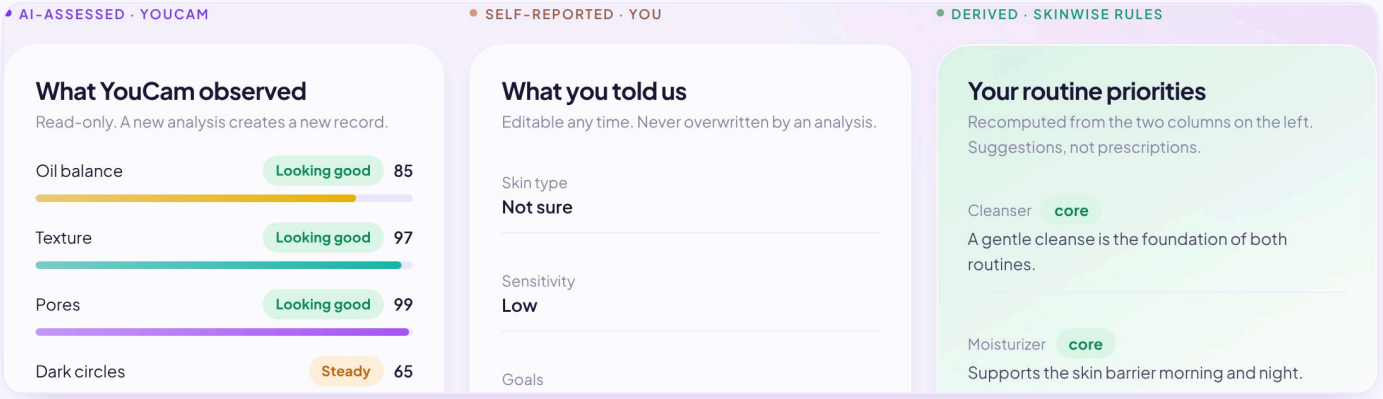
Every YouCam metric as a colourful dial with its own icon and verdict pill, the **AI Score** and **Skin age**, today's routine progress, Skin-to-Shelf coverage and a "For you" row of AI picks for missing steps.

Skin Profile — three sources, never mixed

AI-assessed observations from YouCam (read-only) · **self-reported** skin type, sensitivity, goals, preferences and budget (editable) · **derived** SkinWise priorities with the reason for each.



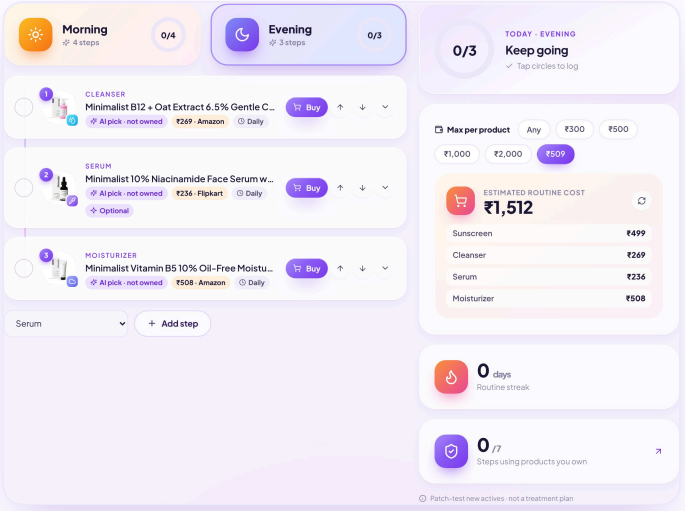
Skin overview — real YouCam scores for the analysed photo, plus YouCam's overall score and skin age.



Skin Profile — AI observations, the user's own goals and SkinWise-derived priorities side by side.

AI-curated AM/PM routine

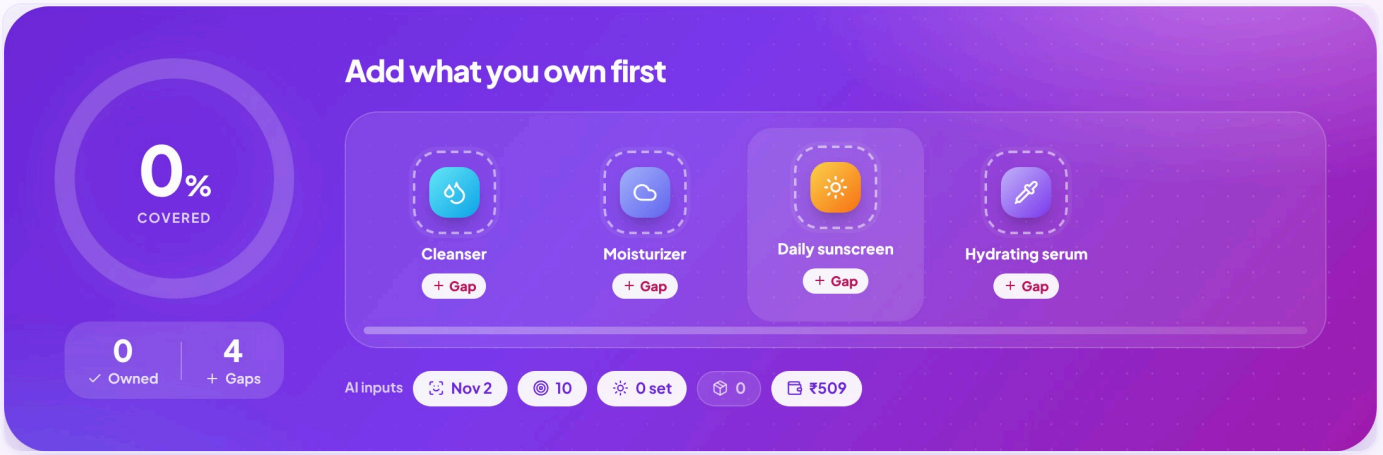
- **Built automatically** from the latest scan, goals and shelf — even when the user owns nothing yet.
- Core steps (cleanser, moisturiser, sunscreen) plus optional steps tied to focus areas and goals; each step explains **why** it exists.
- Steps the user doesn't own get a **real product pick** with photo, price, retailer and a **Buy** link, clearly marked **"AI pick · not owned"**.
- Add, remove, reorder, reassign, mark optional; tick steps daily and build a **routine streak**.
- **Estimated routine cost** and a budget selector sit beside the ritual.



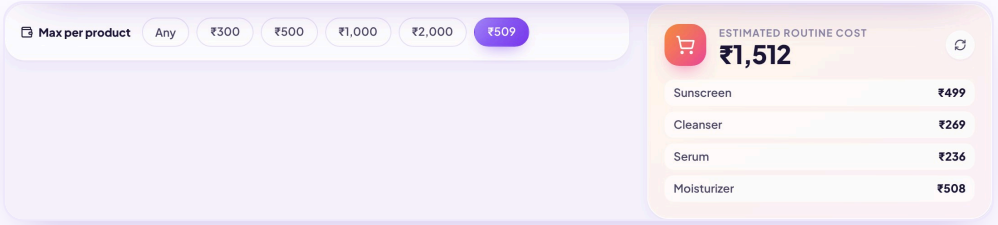
FEATURE 5 · SIGNATURE DIFFERENTIATOR

Smart Skin-to-Shelf engine

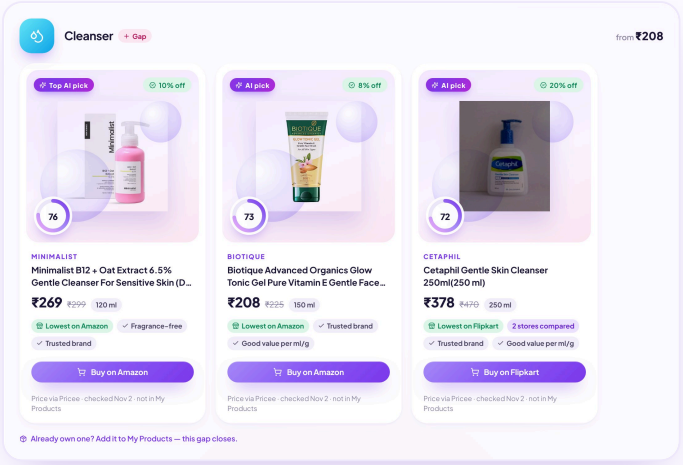
Instead of a shopping list after analysis, SkinWise first checks what the user already owns, identifies **genuine routine gaps**, and only then recommends real products — filtered by budget and priced at the lowest live price across stores.



The visual shelf — coverage ring, owned vs gap per routine step, and the AI inputs used (scan, goals, routine, shelf, budget).



Budget selector (max per product) and estimated routine cost — owned steps count as ₹0.



Recommendation pipeline

- **Needs** come from the YouCam focus areas, the user's goals and the routine (with ingredient keywords such as hyaluronic, ceramide, niacinamide).
- **Real catalog:** in-stock listings via **Pricee**; combos, kits, minis, baby and unrelated products are filtered out.
- **Budget filter:** only products at or under the chosen price; if none fit, the app says so and offers "show any price".
- **Lowest price:** the exact same product (same brand and size) is compared across stores — retailer, ₹ price and a **direct link to that listing**.
- **Never owned automatically:** picks are not added to My Products; already-owned items are never recommended.
- If every step is covered: **"You don't need to buy anything."**

Transparent ranking

50%	25%	25%
Skin suitability	Value	Trusted brand

Value uses price per ml/g to avoid rewarding trial sizes. Missing data is skipped, never guessed.

WHY IT MATTERS

No duplicate buys

Owned products close gaps first, so users stop re-buying what's already on their shelf.

Budget-first

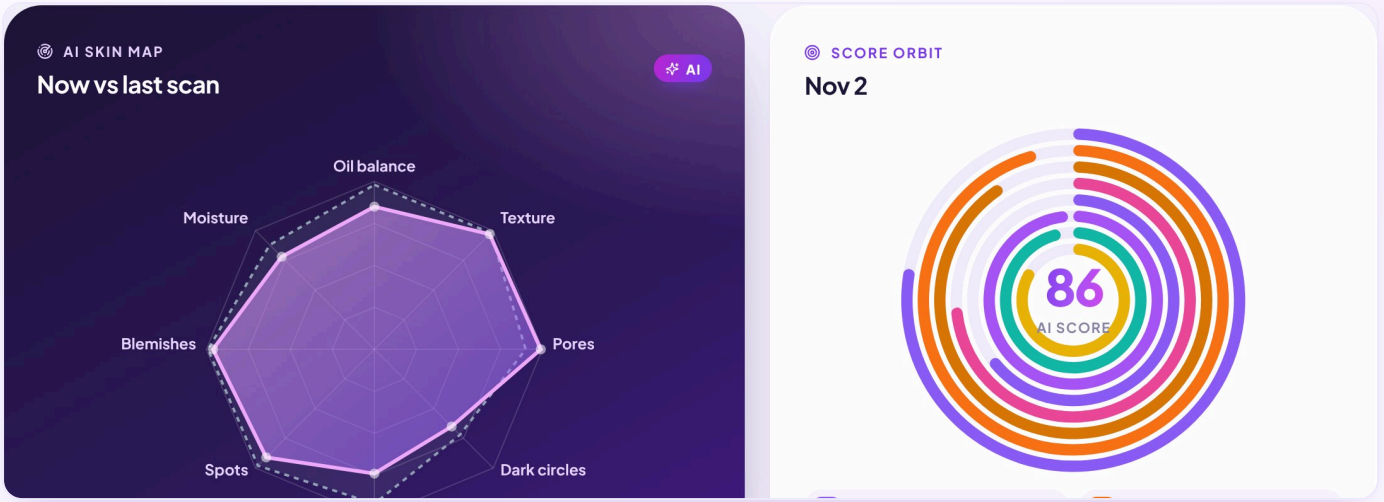
Max-price filter, value per ml/g and a total routine cost make affordability visible before checkout.

Exact listings

Each pick links to the precise product page at the cheapest store — retailer-ready, conversion-ready traffic.

FEATURES 6–9 · JOURNEY, DISCOVER, MY PRODUCTS, ASSISTANT

Track progress, honestly



Skin Journey — AI skin map (radar, now vs previous scan) and score orbit (one ring per concern).

Skin Journey

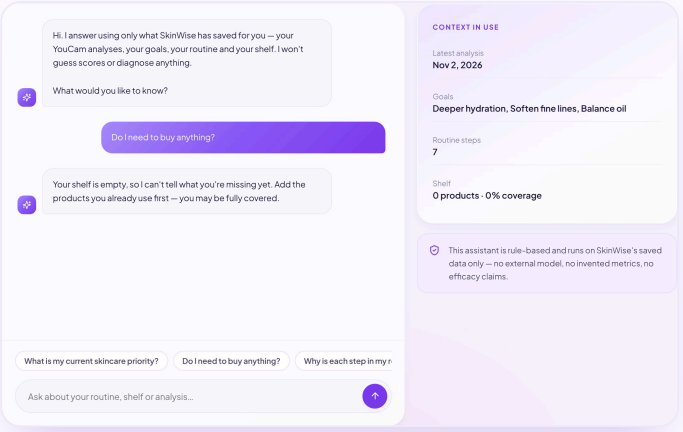
- Every completed scan is saved with a date; stat tiles show scans, latest AI score, best metric and active days.
- Trend charts with **Great / Fair / Focus** zones, before → after bars per concern and a 14-day consistency grid.
- Only metrics both scans returned are compared, and the app never claims a product caused a change.

Discover & My Products

- Search a real product catalog (Open Beauty Facts) with genuine photos; missing data is labelled.
- **My Products** holds only what the user owns (barcode-verified, with role, AM/PM use, expiry and notes), kept separate from AI picks.

AI Skin Coach & Settings

- Answers from the user's own scans, routine and shelf ("Do I need to buy anything?", "What changed?"); declines medical questions.
- Plain-language privacy page, connection status and one-click deletion of analysis records.



AI Skin Coach — grounded in saved data, facts separated from suggestions.

Architecture & stack

LAYER	TECHNOLOGY
Frontend	React 19 · TypeScript · Vite · Recharts · Lucide icons
Backend	Express (Node.js) · Multer (in-memory upload) · JSON persistence + disk cache
AI	YouCam Skin Analysis API v2 (server-side)
Commerce	Pricee live ₹ prices · Open Beauty Facts catalog

Consumer value: real AI insight in plain language, a routine people can follow, and less money spent — buy only what is missing, at the lowest live price, within budget.

Retail value: high-intent, need-matched shoppers; direct links to exact listings; fewer returns; and trust that builds loyalty. Next: YouCam virtual try-on, ingredient-level matching, partner catalogs and accounts.